INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **BCA** Semester: **Second**

Stream: **BCA**

Paper Code: **BCACC201**

Paper Name: **Data Structures and Algorithms**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. What is the purpose of a header file in C programming? **2**

2. What is a primitive data type? Write an example. **2**

3. What is time-space tradeoff in an algorithm? **2**

4. Write two limitations of using arrays in C. **2**

5. Write the advantage of using a stack to implement a non-recursive traversal of a Binary Tree. **2**

6. Explain the key property that differentiates a Binary Search Tree from a Binary Tree. **2**

7. What is the purpose of using a pointer with arrays in C? **2**

8. How can you pass an array to a function in C? **2**

9. What happens if you access an array element out of its bounds? **2**

10. What are the types of rotations used in an AVL tree? **2**

11. Name two comparison-based sorting algorithms that run in O(n log n) time in the worst case. **2**

12. Write the average-case time complexity and the worst-case time complexity for binary search in a sorted array. 2

13. What is the purpose of a good hash function? 2

14. Write the time complexity to print all elements stored in a binary search tree in non-decreasing order. 2

15. How many left shift operations are needed to compute the value of $32n$ from a given integer n without using any multiplication? 2

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Write a C program to print all positive even numbers less than 2025. 5

17. Write a C program to remove duplicate nodes from a linked list in which the elements are stored in non-decreasing order. 5

18. Write a C code for in-order traversal of a binary tree. 5

19. Describe the steps for deleting a leaf node and its parent node in a binary tree when the parent has no other child. 5

20. What are Hash Collisions? Explain two Collision Handling techniques. 5

21. Write in C the declaration of node structure of a linked list. Each node will contain an integer as the element along with a pointer to the next node. 5

22. Write a C program to find the second smallest element in an unsorted array containing 10 distinct integers. 5

23. A binary tree T contains integer elements. Write a C program to count the leaf nodes in T whose elements are less than the elements of their respective parents. 5

24. Write a C program to print the value of $n!$ when n is given as input. 5

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Write a program in C to find the LCM of two positive integers using recursion. Recall that the LCM of two positive integers x and y is the smallest positive 10

integer divisible by both x and y.

26. Explain with a diagram how a node is inserted at the head of a linked list. **10**

27. Draw the binary search tree when the following nodes are inserted one by one: 7, 4, 9, 1, 8, 0, 2, 5, 6, 3. Assume that the tree was initially empty. **10**

28. Give brief introductions of the following: **10**
(a) Division remainder hash function.
(b) Folding method.
(c) Mid square hash function.
(d) Perfect hash function.

29. Given an unsorted array A of n elements and a target sum x, find a pair of elements in the array that adds up to x. If no such pair exists, return -1. For example, if A = {9, 5, 16, 15, 2, 7} and $x = 20$, the output will be (5, 15). **10**

30. Write a program that takes n positive integers as input and computes the sum of their first digits and the sum of their last digits. You should not use array or math.h. **10**

Examples

```
Enter n: 3
Enter the numbers: 7 3 6
Answer = 16, 16.

Enter n: 4
Enter the numbers: 7 30 916 1023
Answer = 20, 16.
```

31. Given the value of n as input, print a right-angled equilateral triangle of height n filled up with *. The outputs for $n = 4$ and $n = 5$ are shown below. **10**

```
* * * *
* * *
* *
*
```

```
* * * * *
* * * *
* * *
* *
*
```

32. Compute the addition of two matrices with dimensions 5-by-7 each, store it in a new matrix, and print its elements on the terminal. **10**

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